Fiscal Year:	FY 2011	**Task Last Updated:**	FY 08/05/2011
PI Name:	Young, Laurence R. Sc.D.		
Project Title:	Sensorimotor Displays and Controls to Enhance the Safety of Human/Machine Cooperation During Lunar Landing		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02139-4301	**Congressional District:**	8
Comments:	Deceased as of August 2021.		
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Crew Health NNJ07ZSA002N
Start Date:	07/01/2008	**End Date:**	09/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date change to 9/30/2012 (from 7/31/2012) per NSBRI (Ed., 1/24/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Oman, Charles (Massachusetts Institute of Technology) Wood, Scott (NASA Johnson Space Center) Duda, Kevin (The Charles Stark Draper Laboratory, Inc.) Estrada, Arthur (US Army Aeromedical Research Laboratory)		
Grant/Contract No.:	NCC 9-58-SA01604		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Landing on any non-terrestrial surface, whether the moon, Mars, Phobos, or an asteroid, depends on the selection and identification of an appropriate location that is level and free of hazards, along with a stable controlled descent to the surface. During crewed landings, astronauts are expected to interact with automated systems, based upon improved terrain maps and sensor updates, to perform tasks such as manual re-designation of landing point, adjustment of descent trajectory or direct manual control. However, sensorimotor limitations, both vestibular and visual, are likely to interfere with performance and safety. This integrated project examines the nature of the anticipated spatial disorientation and terrain perception limits as they affect the transition from automatic to manual control, and develops advanced display countermeasures to overcome these limitations. There are four specific aims investigated in this multi-institution effort: (1) Examine the nature of anticipated sensorimotor difficulties (e.g., spatial disorientation, limits on terrain perception) as they affect the transition from automatic to manual control, (2) Develop and evaluate advanced display countermeasures for enhancing situation and terrain awareness and for overcoming performance limitations caused by reduced visibility associated with lunar lighting, terrain reflectivity and the absence of atmosphere utilizing Draper Laboratory's fixed-base lunar lander cockpit simulator for full human-in-the-loop evaluation. (3) Evaluate the effectiveness of the cockpit displays during human-in-the-loop manual control in the JSC TTS during "critical" and "hover" tasks testing the tilt-translation and tilt-gain illusions of altered acceleration sensitivity as it applies to lunar gravity following a period of weightlessness, and (4) Perform a series of evaluations of the displays using the U.S. Army Aeromedical Research Laboratory's six-degree-of-freedom helicopter simulator as a lunar landing analog for replicating lunar lighting and the various parameters associated with dust "brownout" conditions. Although the specific target for the original research was lunar landing, the methods and results are equally applicable to landing on any planetary or asteroid body that has a gravity different than one, and in which dust blowback is a potential challenge. In our third year of work, we continued to make strides on Aims 1 and 2, while beginning work on Aims 3 and 4. For Aim 1, further analysis was completed on the experiment done in collaboration with Dr. Bilimoria's group at NASA Ames Research Center. The experiment found that subjects reported misperceptions of vehicle orientation and horizontal velocity, even in the case where they were provided with a simulated out-the-window view. This was likely due to the small-field-of-view forward facing window which limited their complete perception. In Aim 2, we have developed an experimental testbed for lunar landing final approach and terminal descent in the Draper fixed-base simulator. This testbed simulation was used to test novel display prototypes including an achievability contour display, which provides the pilot information on which areas on the lunar surface are achievable given the remaining amount of fuel. In addition to the novel displays, a complete primary flight display (PFD) has been developed for simulation purposes. An experiment using these displays to land on the lunar surface was carried out in the Draper simulator. The achievability contour display was seen to improve subjective situation awareness and workload while improving the pilot's confidence in making landing point redesignations with a limited amount of fuel as compared to the auditory display type used during Apollo. Advanced blended control modes have been developed at Draper to assist the astronauts in the different phases of lunar landing. For Aim 3, the Draper and MIT group's display prototypes have been implemented in the NASA JSC Tilt-Translation Sled (TTS). The TTS will be used to simulate the sensorimotor challenges astronauts are expected to face during a lunar landing. In particular, the Critical Tracking Task (CTT) was implemented in the TTS and will be used to evaluate the advanced display countermeasures for their ability to improve pilot performance. For Aim 4, the US Army 6-DOF helicopter simulator has been modified to incorporate the properties of lunar dust blowback. An experiment has been designed to study the impact on dust blowback on pilot perceptions and landing performance. For the fourth year, we will continue our advance display countermeasure development in the Draper fixed based simulator. These display prototypes will be tested in future experiments for their ability to improve pilot perceptions and performance during lunar landing. The NASA JSC TTS experiment will be completed, determining the effect of the sensorimotor challenges expected during lunar landing on pilot performance as well as the improvement expected using the display countermeasures. The Army 6-DOF motion simulator experiment will be completed to study the effect of dust blowback on astronaut perceptions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our goal is to determine the limits of human performance under likely extra-terrestrial landing conditions that may cause spatial disorientation. This contributes to a better understanding of visual and vestibular conditions contributing to spatial disorientation during landing and the resulting effects on human manual control. We will have demonstrated display and control system interfaces to reduce pilot workload, improve situation awareness, and mitigate spatial disorientation to ensure a safe crewed lunar landing. Finally, the work may also have terrestrial applications in mitigating the risk of helicopter accidents by suggesting new techniques to address problems associated with brownout during landing. In particular the achievability contour display being prototyped may have applications in helicopter flight planning, coping with brownout, as well as mission management aspects for guiding air-drops to locations given current environmental conditions (e.g. wind, air density). This display concept can be used for any energy-constrained terrestrial approach or landing. The Observer vestibular-visual spatial orientation model has been used to predict astronaut perceptions during lunar landing motions. This work in applying this model to vehicle trajectories has applications for terrestrial vehicles as well. The vehicle motions experienced during terrestrial accidents can be simulated in the model to determine if spatial disorientation may have played a role. Advances in the understanding of the role of motion cues and visual surround will contribute to aviator training relevant to spatial disorientation.
	We have extracted relevant parameters from candidate trajectories within the NASA Autonomous Landing and Hazard Avoidance Technology (ALHAT) Project trade space. These parameters were used as inputs to the Observer Model, which is a physiologically based model of spatial orientation, to estimate perceived orientation of the astronaut within the lunar landing vehicle. Predictions of pilot perceptions were seen to differ from the actual vehicle orientation. The Observer model results were partially validated with a human subject experiment in the NASA-Ames Vertical Motion Simulator (VMS). In this experiment, subjects reported orientation perceptions during landings. Subjects indicated perceptions that differed substantially from the actual vehicle state, even when provided with a simulated out-the-window view of the lunar surface. Only when provided with head-down state displays did perceptions improve. An experimental testbed for piloted lunar landing has been developed in Draper's fixed base simulator. Prototype control modes and primary flight displays that were implemented in MATLAB/Simulink for initial evaluation have been expanded upon and implemented in the Draper simulator. In addition to primary flight displays, novel display concepts have been developed. In particular an achievability contour display, which provides the pilot information on achievable landing areas given the current fuel level, has been developed and implemented in the simulator for experimentation. It

Task Progress:	was seen that subjective workload and situational awareness improved when using the achievability contour display. Furthermore, when using the achievability contour display subjects were more likely to redesignate to preferable landing points even if they required more fuel to be reached. We were able to enlist the expert evaluations of four Apollo astronauts, resulting in significant modification of the proposed display. The protocol for the Aim 3 experiment at Johnson Space Center (JSC) using the Tilt-Translation Sled (TTS) has been reviewed and approved by the JSC-IRB. The display prototypes developed at Draper have been integrated into the TTS, including a conventional primary flight display and a simulated out-the-window-view. The TTS has been modified to have these displays fixed to the subject's chair. A critical tracking task (CTT) of increasing difficulty has been implemented in the TTS. The US Army Aeromedical Research Lab (USAARL) has augmented the dust blowback simulation currently in the 6-DOF helicopter simulator to conform to what is expected during lunar landings with regolith blowback in terms of particle size, height of dust onset, etc. The primary flight display developed at MIT/Draper and lunar landing-like trajectories have been implemented. The experimental protocol is under review by the USAARL-IRB.
Bibliography Type:	Description: (Last Updated: 01/21/2026)
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